

1. Introduction

Although recent deep learning models are effective in food image classification, it is common practice to only identify a dish. Nutritional labels are not typically included on many food items particularly the ones eaten outside or cooked at home and therefore individuals can hardly estimate the calories or the macronutrient content.

Individuals who are interested in losing weight or adopting a balanced diet tend to make their food choices based on the visual impression of the food even though the nutritional content of similar food items is significantly different. The automatic estimation of the nutritional information based on food pictures can be beneficial even when the estimate is not accurate.

In this project, food recognition is proposed as a multi-task learning problem, so we concurrently solve the problems of food classification and nutrition prediction.

This project combines two complementary open datasets from Kaggle: the **Food-101 dataset** and the **Food-101 Nutritional Information dataset**. The **Food-101 dataset** consists of 101 food categories, each containing 1,000 images collected from real-world sources. It is widely used as a benchmark for food image classification due to its visual diversity and challenging inter-class similarities. The standardized train–test split and balanced class distribution ensure fair evaluation and reliable comparison. The **Food-101 Nutritional Information dataset** provides nutritional statistics for the same 101 food categories enabling a straightforward class-level integration between images and nutrition labels. For each food class six nutritional attributes were selected for prediction: **calories, protein, fat, carbohydrates, sugars, and sodium**.

The convolutional neural network that uses a pretrained ResNet-18 base is trained to give predictions of both the food category and six nutritional characteristics of 100-gram of food. By sharing a common visual backbone, the model learns representations that are informative for both tasks. Besides this, a creative extension is also investigated where the space of learned embedding is applied to suggest food alternatives of similar appearance but lower calorie content, which proposes a practical usage for the model.

This project has the following objectives: Overcome the inaccessibility of nutrition information about most food products; Assess the efficiency of multi-task learning in classification of foods and nutrition estimation; Show the extent to which learned visual representations can benefit health-aware decision making.

2. Methodology

2.1 Preprocessing

Combined the **Food-101 dataset** and the **Food-101 Nutritional Information datasets** used **calories, protein, fat, carbohydrates, sugars, and sodium** as nutritional attributes. The dataset provides a standard split of 750 training images and 250 test images per class. Since nutritional values in the dataset are reported per serving, all nutrition values were converted to **per-100-gram units** to ensure consistency across food categories. When multiple nutrition entries existed for a given food class, values were averaged to obtain a single representative nutrition vector per class. Minor naming inconsistencies between the two datasets were resolved using manual aliases and string matching.

For training stability, nutrition targets were **z-score normalized** using the mean and standard deviation computed across all 101 food classes. These normalization statistics were stored and later used to convert predictions back to real units during evaluation.

Image preprocessing followed standard practice for pretrained convolutional networks. All images were resized to **224×224 pixels**. During training, random horizontal flipping was applied as data augmentation, while only resizing was used for the test set. Images were normalized using ImageNet mean and standard deviation values to match the pretrained ResNet backbone.

2.2 Model Architecture

The model follows a multi-task learning architecture with a shared visual backbone and two task-specific output heads.

A ResNet-18 pretrained on ImageNet was used as the backbone feature extractor. The final fully connected layer of ResNet-18 was removed, producing a 512-dimensional feature vector for each input image. This shared representation serves as the input to both prediction heads.

Two parallel linear heads were attached to the backbone: A classification head, which outputs logits for the 101 food classes; A regression head, which outputs six continuous values corresponding to calories, protein, fat, carbohydrates, sugars, and sodium per 100g of food.

No activation function was applied at the output layers. For classification, raw logits are required because softmax is applied internally by the cross-entropy loss. For regression, linear outputs are appropriate for predicting real-valued quantities.

2.3 Loss Functions and Optimization

The model was trained using a combined multi-task loss consisting of:
Cross-entropy loss for food classification;
Mean squared error (MSE) for nutrition regression.

The total loss is defined as: $L = L_{cls} + \lambda L_{reg}$

where $\lambda=0.15$ controls the relative importance of the regression task. This weighting prioritizes classification accuracy while still encouraging meaningful nutrition predictions.

Optimization was performed using the Adam optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} for regularization. The model was trained for 6 epochs, which was sufficient to reach stable performance without excessive overfitting.

2.4 Training and Evaluation Procedure

Training was performed using mini-batches of size 32 for the training set and 64 for the test set. During each epoch, the model parameters were updated using backpropagation on the combined loss. Classification accuracy was computed using the top-1 predicted class.

Evaluation was conducted after each epoch on the test set with gradients disabled. The best model was selected based on highest test accuracy, which occurred at epoch 6.

To provide a comprehensive evaluation, multiple metrics were used: Top-1 and Top-5 accuracy for classification; Per-class accuracy and confusion matrix analysis to identify common failure modes; Mean Absolute Error (MAE) and R^2 score for each nutritional attribute; Scatter plots comparing predicted and ground-truth calorie values.

2.5 Qualitative Analysis

In addition to quantitative metrics, qualitative evaluation was performed using randomly selected, class-diverse test images. For each example, the predicted class and predicted nutrition values were compared against ground truth. This analysis highlights cases where the model produces reasonable nutritional estimates even when classification is incorrect, demonstrating the benefit of multi-task learning.

2.6 Creative Extension: Similarity-Based Lower-Calorie Recommendation

As an optional creative component, the learned embedding space was used to build a health-aware food recommendation system.

Class-level visual prototypes were constructed by averaging normalized backbone embeddings from multiple training images per class. Given a query image, cosine similarity between its embedding and all class prototypes was computed to identify visually similar foods.

Among the most similar classes, candidates with significantly lower calorie density (per 100g) were selected as recommendations. This approach allows the model to suggest visually similar but lower-calorie alternatives, demonstrating a practical downstream application beyond classification and regression.

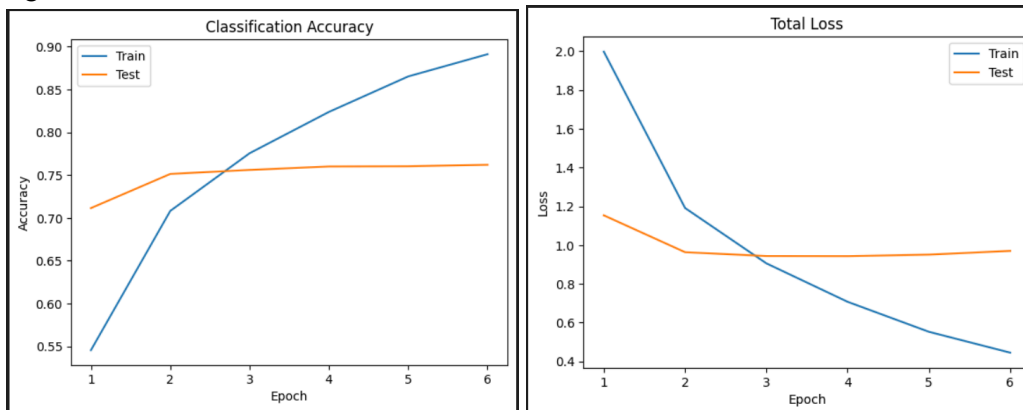
2.7 Implementation Details

The model was implemented in PyTorch and trained using GPU. All experiments were conducted in a reproducible manner, with fixed preprocessing steps and saved model checkpoints, including normalization statistics and training history.

3. Results & Analysis

3.1 Training Behaviour

The model was trained for 6 epochs using a multi-task objective combining classification and nutrition regression losses.



Classification accuracy on the training set increased steadily from **54.6% to 89.1%**, indicating effective feature learning. Test accuracy improved rapidly during the first two epochs and then plateaued, reaching a maximum of **76.21%** at epoch 6. The gap between training and test accuracy in later epochs suggests mild overfitting, which is expected given the limited number of training epochs and the model's capacity.

Similarly, the total training loss decreased monotonically from **1.997 to 0.445**, while test loss decreased initially and then stabilized around **0.97**. This behaviour indicates stable optimization without divergence. The best-performing model was selected based on **highest test accuracy**, achieved at **epoch 6**.

3.2 Classification Performance

The final model achieved: **Top-1 Accuracy: 76.21%; Top-5 Accuracy: 93.41%**

The significantly higher Top-5 accuracy indicates that, even when the model's top prediction is incorrect, the correct class is usually among the top visually plausible alternatives. This is particularly important for food recognition, where many dishes share similar visual characteristics.

Per-Class Accuracy and Failure Cases

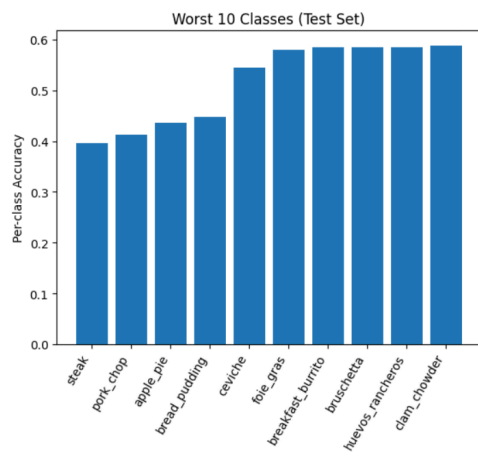


Figure shows the **10 worst-performing classes** by per-class accuracy.

These classes often exhibit **high intra-class variability** and **strong visual overlap** with similar dishes, making them harder to distinguish using image-only cues.

Confusion Analysis

Analysis of the confusion matrix revealed that most misclassifications occurred between **semantically and visually similar foods**.

```
Top confusion pairs (true -> predicted):
steak -> filet_mignon : 84
clam_chowder -> lobster_bisque : 61
pork_chop -> filet_mignon : 44
chocolate_cake -> chocolate_mousse : 41
tuna_tartare -> beef_tartare : 36
hamburger -> pulled_pork_sandwich : 33
poutine -> french_fries : 29
scallops -> crab_cakes : 27
spaghetti_carbonara -> spaghetti_bolognese : 27
falafel -> crab_cakes : 27
```

These errors are reasonable given shared ingredients, textures, and presentation styles. Importantly, many such confusions remain acceptable in downstream applications such as recommendation systems.

3.3 Nutrition Regression Performance

The regression head predicts six nutritional values per 100g. Performance was evaluated using **Mean Absolute Error (MAE)** and **R² score** in real units.

Regression MAE (per nutrient, per 100g):	Regression R ² (per nutrient):
calories : 35.290	calories : 0.698
protein : 2.504	protein : 0.623
fat : 2.873	fat : 0.583
carbs : 6.398	carbs : 0.657
sugars : 3.501	sugars : 0.725
sodium : 54.150	sodium : 0.516

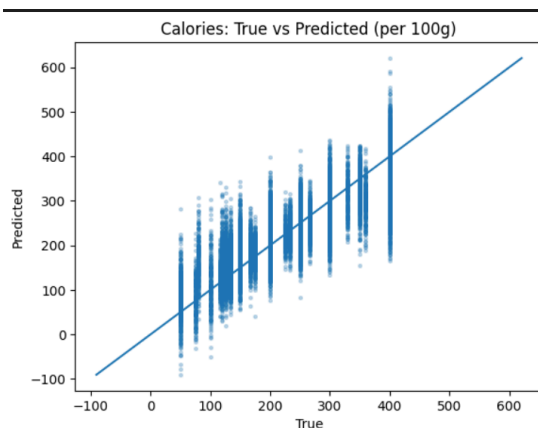
Considering that food calorie values often range from 100–600 kcal per 100g, an average error of ~35 kcal represents a **moderate and practically useful level of accuracy**.

The highest R² was achieved for **sugars and calories**, indicating that visual cues are particularly informative for estimating energy-dense components. Sodium achieved the lowest R², likely because salt content is weakly correlated with visual appearance.

Regression Visualization

Figure W shows a scatter plot of **true vs predicted calories per 100g**.

Predictions follow a clear linear trend, although variance increases for higher-calorie foods. This suggests the model captures global calorie structure but struggles with fine-grained differences among similar dishes.



3.4 Qualitative Evaluation

Qualitative examples further illustrate model behaviour. In several cases, the model correctly predicted both class and nutrition values (e.g., *baklava*, *bibimbap*, *beignets*), with predicted nutritional values closely matching ground truth.

In misclassified cases (e.g., *apple_pie* → *samosa*), the predicted nutritional profile remained **reasonable**, reflecting the caloric and macronutrient similarity between the two dishes. This highlights the benefit of multi-task learning: even when classification fails, nutritional estimates may remain informative.

```
=====
True class: baklava | Pred class: baklava
Pred nutr: {'calories': 334.9074401855469, 'protein': 4.306828022003174, 'fat': 15.814107894897461, 'carbs': 42.539791107177734, 'sugars': 26.04132843017578, 'sodium': 172.6382446289}
GT nutr: {'calories': 330.0, 'protein': 5.909523963928223, 'fat': 15.909523963928223, 'carbs': 40.0, 'sugars': 25.409523010253906, 'sodium': 120.0}
=====
True class: beef_carpaccio | Pred class: tuna tartare
Pred nutr: {'calories': 200.05165100097656, 'protein': 18.064014434814453, 'fat': 9.184292793273926, 'carbs': 6.476676940917969, 'sugars': -1.1800799369812012, 'sodium': 299.95693969}
GT nutr: {'calories': 200.0, 'protein': 20.0, 'fat': 12.0, 'carbs': 1.4200000762939453, 'sugars': 0.0, 'sodium': 300.0}
=====
True class: beef_tartare | Pred class: tuna tartare
Pred nutr: {'calories': 227.37319946289062, 'protein': 11.833819389343262, 'fat': 14.253288269042969, 'carbs': 14.820232391357422, 'sugars': 6.545708656311035, 'sodium': 258.44979858}
GT nutr: {'calories': 250.0, 'protein': 25.106666564941406, 'fat': 15.106666564941406, 'carbs': 2.0, 'sugars': 1.419999599456787, 'sodium': 400.0}
=====
True class: beet_salad | Pred class: beet_salad
Pred nutr: {'calories': 107.26416778564453, 'protein': 3.7166833877563477, 'fat': 5.145476341247559, 'carbs': 9.371807098388672, 'sugars': 3.432173252105713, 'sodium': 210.5661621093}
GT nutr: {'calories': 80.0, 'protein': 3.173332691192627, 'fat': 2.0, 'carbs': 12.0, 'sugars': 8.0, 'sodium': 150.0}
=====
True class: beignets | Pred class: beignets
Pred nutr: {'calories': 377.3638000488281, 'protein': 5.598293781280518, 'fat': 19.462289810180664, 'carbs': 43.96722412109375, 'sugars': 20.209928512573242, 'sodium': 242.8298339843}
GT nutr: {'calories': 400.0, 'protein': 5.909523963928223, 'fat': 20.0, 'carbs': 50.0, 'sugars': 24.090476989746094, 'sodium': 200.0}
=====
True class: bibimbap | Pred class: bibimbap
Pred nutr: {'calories': 143.52694702148438, 'protein': 6.762770652770996, 'fat': 4.416009426116943, 'carbs': 16.34968376159668, 'sugars': 2.7333526611328125, 'sodium': 271.5529174804}
GT nutr: {'calories': 150.0, 'protein': 5.0, 'fat': 3.999999523162842, 'carbs': 20.0, 'sugars': 3.3204760551452637, 'sodium': 266.6795349121094}
```

3.5 Creative Extension: Health-Aware Food Recommendation

As a creative extension, the learned embedding space was used to build **class prototypes** and recommend **visually similar but lower-calorie alternatives**.

```

Model predicted: apple_pie | calories/100g: 300.0
Alternatives (visually similar + lower calories):
- grilled_cheese_sandwich  sim=0.761 cal=266.6 drop=33.4
- bread_pudding           sim=0.738 cal=250.0 drop=50.0
- pizza                   sim=0.729 cal=250.0 drop=50.0
- ice_cream               sim=0.702 cal=200.0 drop=100.0
- croque_madame           sim=0.694 cal=225.1 drop=74.9
- hot_dog                 sim=0.691 cal=250.0 drop=50.0
- french_toast            sim=0.683 cal=250.0 drop=50.0
- pancakes                sim=0.666 cal=250.0 drop=50.0

```

These suggestions preserved visual similarity while reducing calorie density, demonstrating a practical application beyond standard classification.

3.6 Summary of Results

Overall, the model achieves strong classification performance on Food-101, produces meaningful nutrition predictions from images, exhibits interpretable failure modes, supports downstream recommendation tasks

4. Conclusion & Future Work

Using the Food-101 dataset and the Food-101 Nutritional Information dataset, the model was trained to recognize food appearance with the semantic class names as well as with the nutritional properties. The experimental findings indicate that the model exhibits high classification accuracy with Top-1 accuracy of 76.21% and Top-5 accuracy of 93.41% while also giving meaningful nutrition predictions with acceptable error rates. The study of confusion patterns shows that the majority of the errors in classification happen between similar dishes in terms of visual and semantic features, which is natural, considering the character of food imagery.

In addition to quantitative performance, qualitative analysis also points to a significant strength of multi-task learning, namely that despite the inaccuracy of classification, the predicted nutritional values tend to be realistic. The creative extension based on embedding-based similarity is yet another indication of the practical potential of the learned representations, which can be used to make health-conscious food suggestions, e.g., by recommending visually similar yet lower in calories food options.

Limitations

Class-level nutrition labels: The nutrition value is determined at the level of classes instead of the image level. This disregards the difference in the quantity of portions, ingredients, and cooking techniques of the same type of food.

Short training time: The model was only trained with a few epochs and this could be a limiting factor in terms of its capacity to converge and generalize.

Visual ambiguity: Certain nutritional qualities, including sodium levels, are loosely associated with color, leading to lower regression performance.

Single backbone architecture: ResNet-18 was the only backbone which can constrain representational capacity of fine-grained food differences.

Future Work

Image-level nutrition labels: With portion-specific or image-level nutrition annotations, more precise and personalized nutrition estimation would be possible with the help of datasets.

Bigger or more powerful backbones: It might be worth looking into more powerful or more complex architectures, like ResNet-50, EfficientNet, or Vision Transformers.

Other modalities: At the expense of nutrition prediction, adding the text description, or the ingredient lists or user input may increase the accuracy of nutrition prediction significantly.

Estimating portion size: Incorporation of the object size estimation and depth information could aid in estimating the actual portion size and not the values per 100 gram.

Front-end applications: The recommendation system might be expanded to include a complete application that helps manage calories, weight loss, or dietary recommendations.